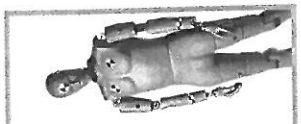
2 Indirect observation of electron shells

Purpose

To observe coloured light produced when electrons jump from one electron shell to another.

Materials

- cotton buds or small cotton balls
- chloride solutions of barium, sodium, copper, potassium and calcium
- 5 small beakers
- 1 large beaker of water
- Bunsen burner
- tongs



SAFETY

Some of the chemicals in this prac are toxic. Make sure you wear safety glasses and gloves when handling chemicals. Avoid contact with skin, mouth and eyes. Wash your hands thoroughly at the end of the practical.

Procedure

1. Copy the table from the results section into your notes.
2. Pour about 2 to 5 mL of the chloride solutions into labelled beakers—one per beaker. Label the beakers so you can identify each solution.
3. Light the Bunsen burner, leaving it on the yellow safety flame until you are ready to insert a saturated cotton bud or ball. Fill another beaker with water.
4. With the tongs pick up the cotton bud or ball and soak it in the barium chloride solution.
5. Place the soaked cotton bud in the blue flame of the Bunsen burner for about 2 seconds and observe the flame colour. Immediately remove the cotton bud or ball and drop it in the beaker of water. Do not let it catch fire. Record the results in your table.
6. Repeat the procedure for the remaining solutions.

Results

Copy the following table into your workbook and complete it.

Solution	Metal element	Observation
Barium chloride		
Sodium chloride		
Copper chloride		
Potassium chloride		
Calcium chloride		

Discussion

1. Different-coloured light is produced depending on whether the electron shells are far apart or close together.
   - When the electron shells are far apart, green, blue or violet light is produced.
   - When the electron shells are close together, red, orange or yellow light is produced.

   Use your results to **classify** the compounds in your experiment as having electron shells that are either 'far apart' or 'close together'.
2. **Propose** what compounds you might use to make green and purple fireworks.